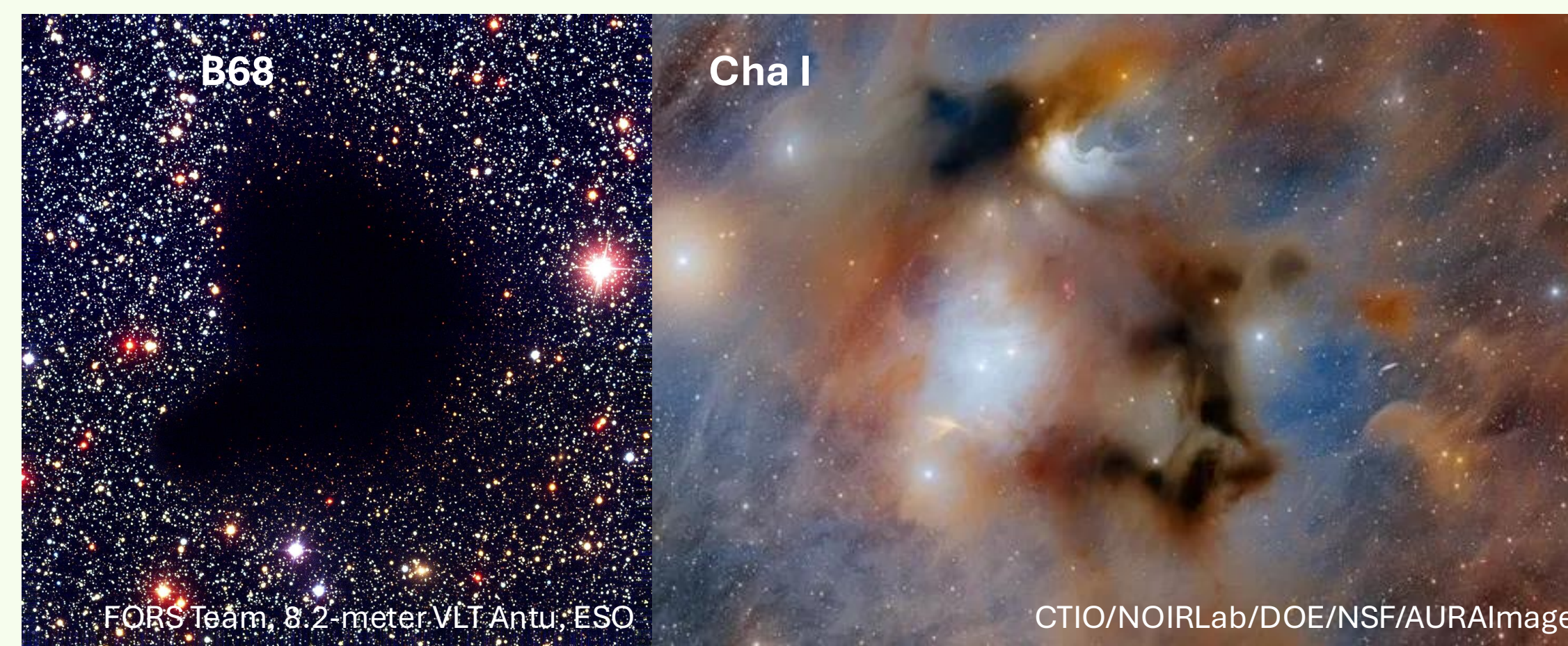


Why do we study molecular cloud ices?

Molecular clouds are the cradle of *stars*, precursors to the protostars and protoplanets. The elemental and molecular *budget* of stars and planets is tied to that of the molecular clouds that came before.

The **icy** mantles on the clouds' dust grains are the largest molecular reservoir in the ISM, and can only be observed in molecular clouds (cold and dense). These are the building blocks of **life**.

Common ices: H₂O, CO₂, CO, CH₄, NH₃, CH₃OH



How do we study molecular cloud ices?

The ices are observed as absorption features against a background continuum.

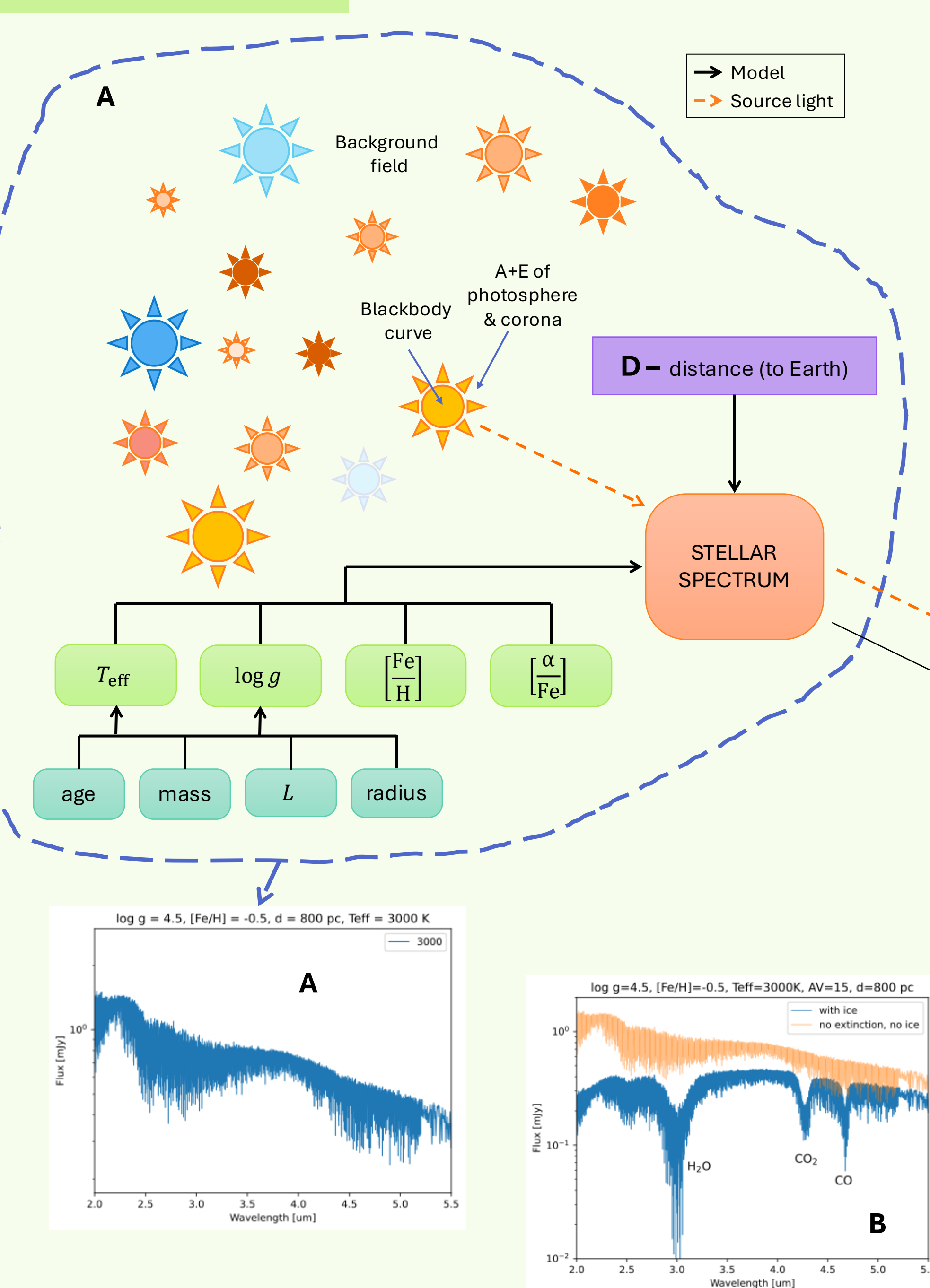
They are measured as *optical depth*:

$$\tau_\nu = -\ln \frac{F_{\text{observed}}(\nu)}{F_{\text{continuum}}(\nu)}$$

and converted to column density N (A the band strength):

$$N = \int \tau_\nu d\nu / A$$

METHODOLOGY



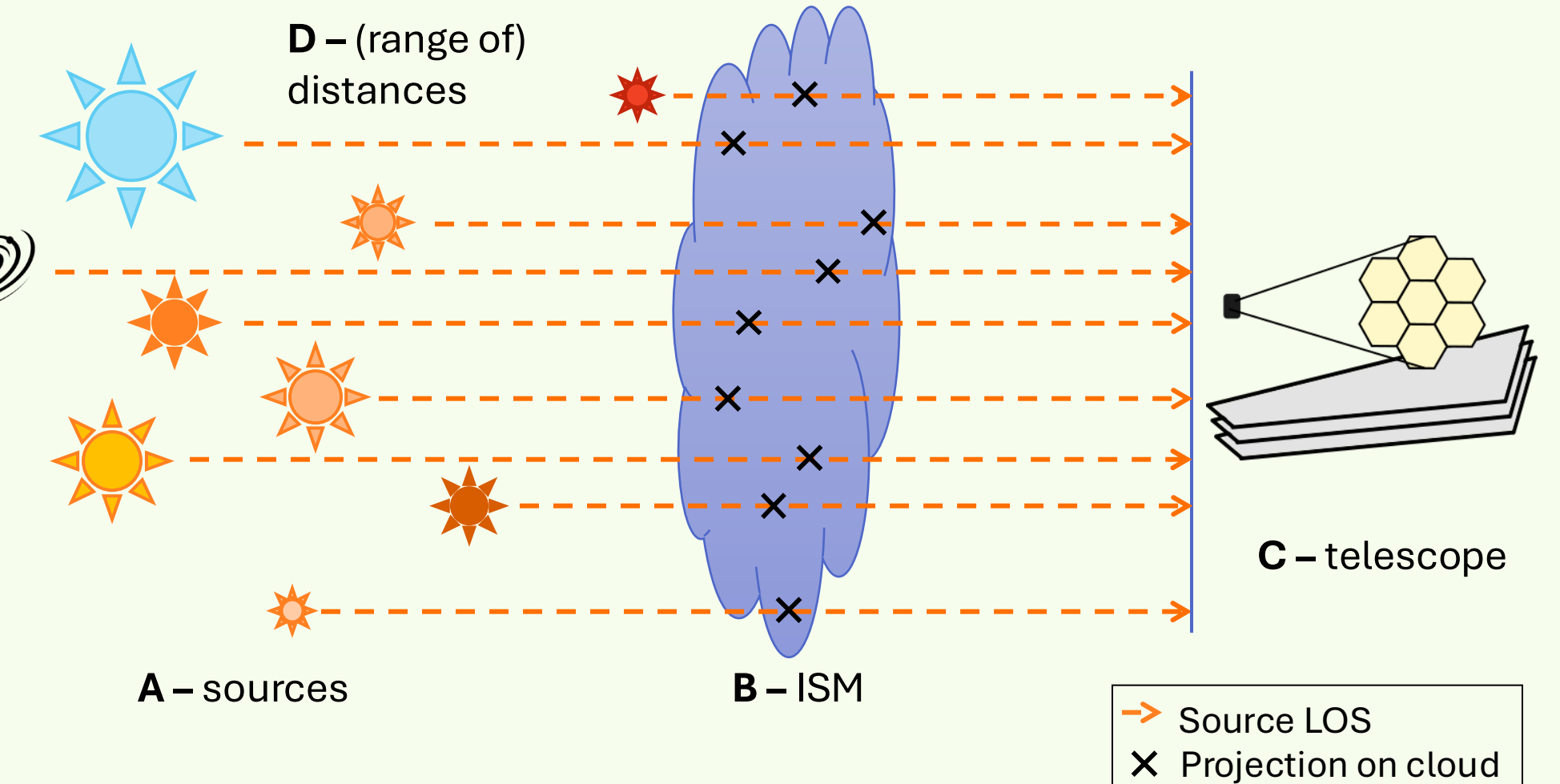
ICE MAPPING

It's a technique to study the spatial *extent* of molecular clouds by concurrent observation of multiple LOS (lines of sight) probing the molecular cloud.

Ices are seen as absorption features against a continuum emission from background sources [2].

BUT! The photospheric lines from the sources, as well as gas and dust effects make it difficult to *qualify* and *quantify* the ices

→ need to accurately *baseline* the continuum behind the ices



BASELINING

It's an *inverse* problem: from the observation(s) I need to extract the variables/parameters that lead to it.

The problem is that many parameters are highly **degenerate**, and different combinations may produce similar spectra:

- **T_{eff} , distance, extinction:** all change the flux intensity and the slope of the spectrum
- **Metallicity, $\log g$ and T_{eff} :** change the type and intensity of photospheric lines in the spectrum

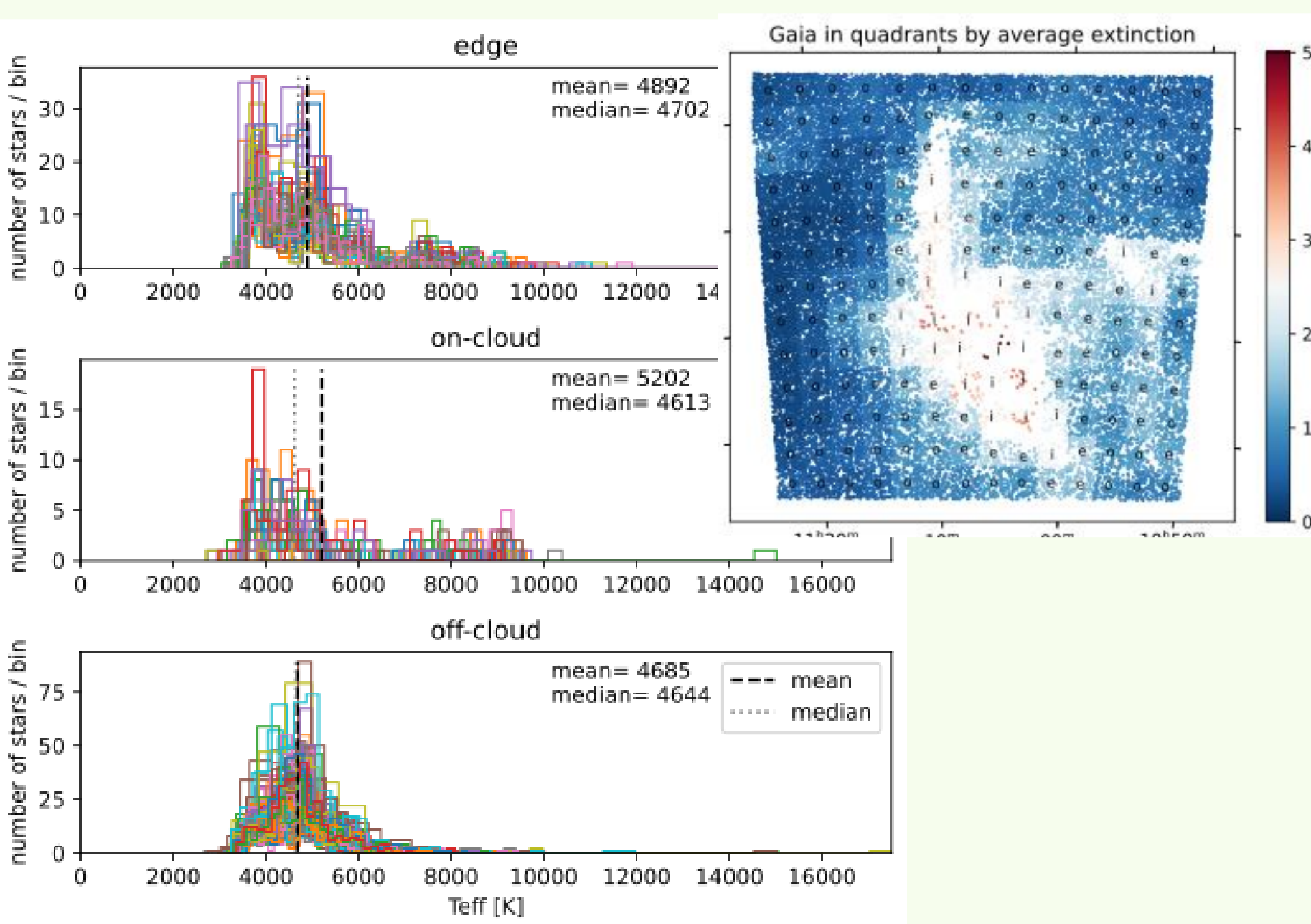
The issue is exacerbated when only parts of the spectrum are visible, due to either saturation, noise or out-of-coverage spectral traces. Less datapoints implies less constraints, making the resolution of degeneracies worse.

RESULTS

Understanding the background population

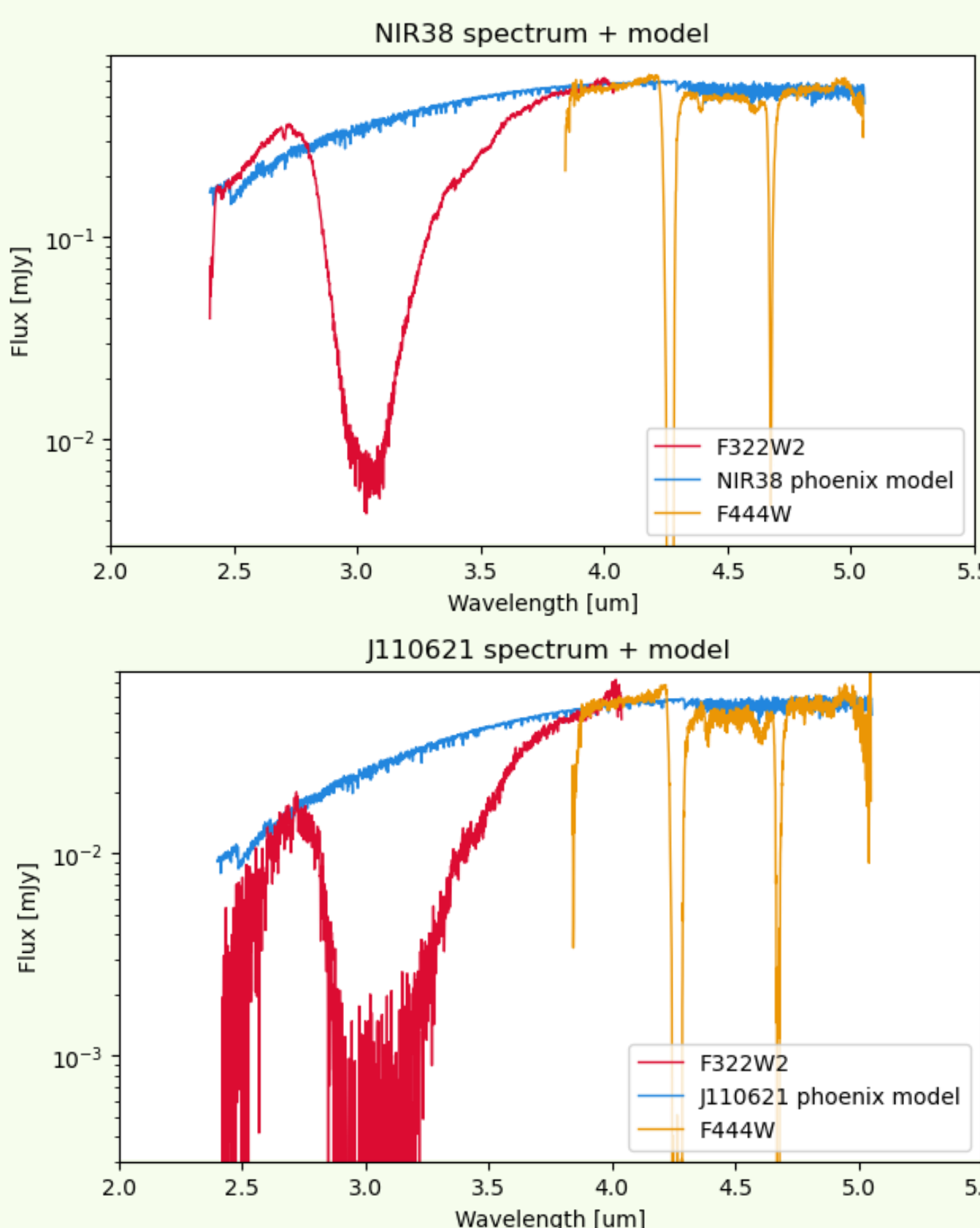
Cataloguing stars using Gaia DR3 to determine the stellar population distribution in the background field. Divided area around molecular cloud in smaller chunks, and for each computed the T_{eff} histogram of stars within.

- On, off and edge of the cloud classification approximated by average extinction in G band (Gaia DR3)
- Distribution consistent across the field, median $T_{\text{eff}} \sim 4600$ K.
- On-cloud has a second peak at ~ 8000 K, due to observational bias (only see bright sources through the cloud).



Can I reproduce observations?

As proof of concept, I manually tuned the parameters to simulate the observations of the stars NIR38 and J110621 in the Cha I molecular cloud [1][2].

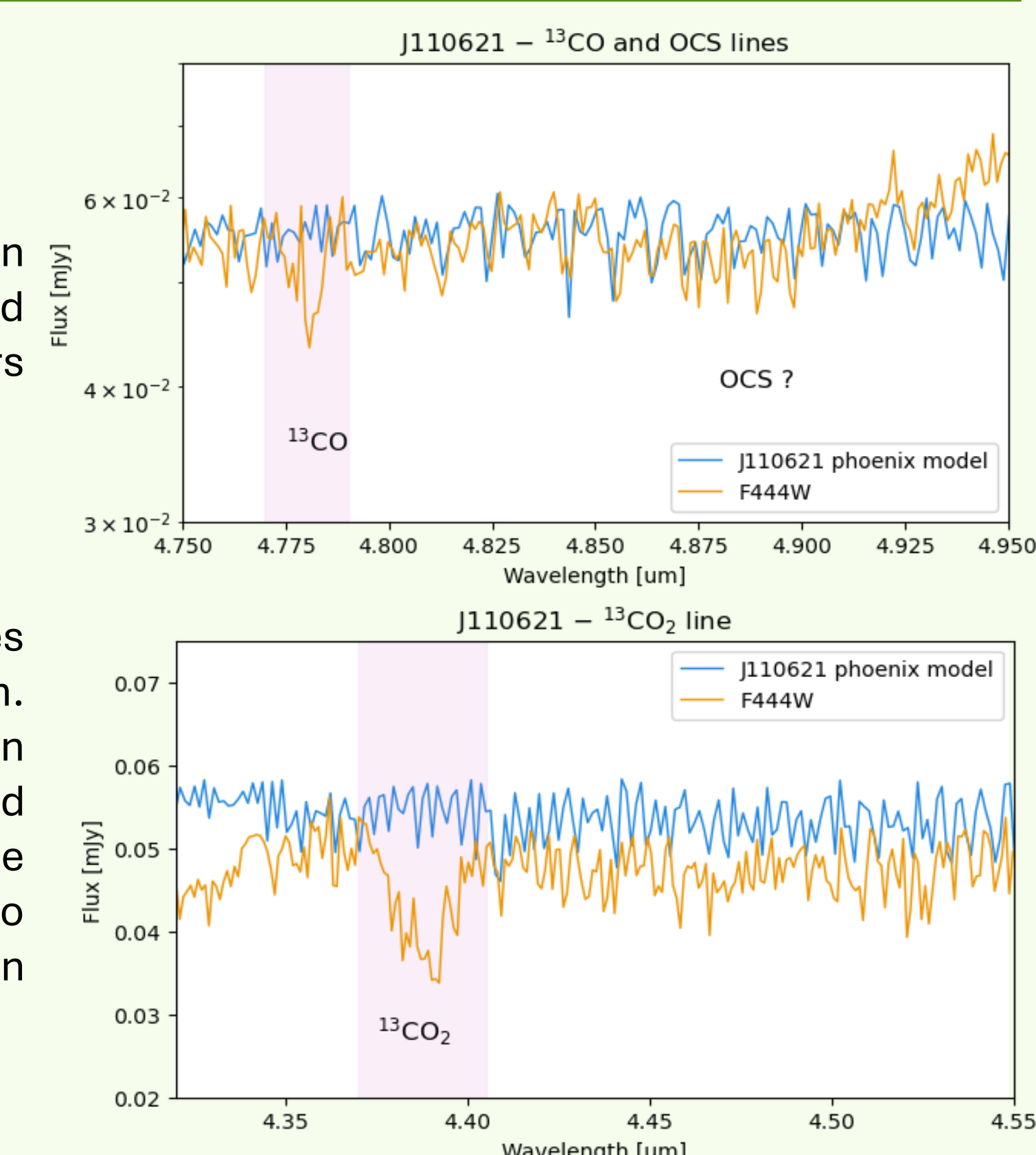


Right:

Comparison between observed spectra and simulated continuum for stars NIR38 and J110621. (data from ICEAGE [1])

Left:

Zoom-in on minor ice features present in J110621 spectrum. Note the good match between simulated and observed photospheric lines in the continuum. Contrary to [1], no OCS can be detected using an accurate baseline.



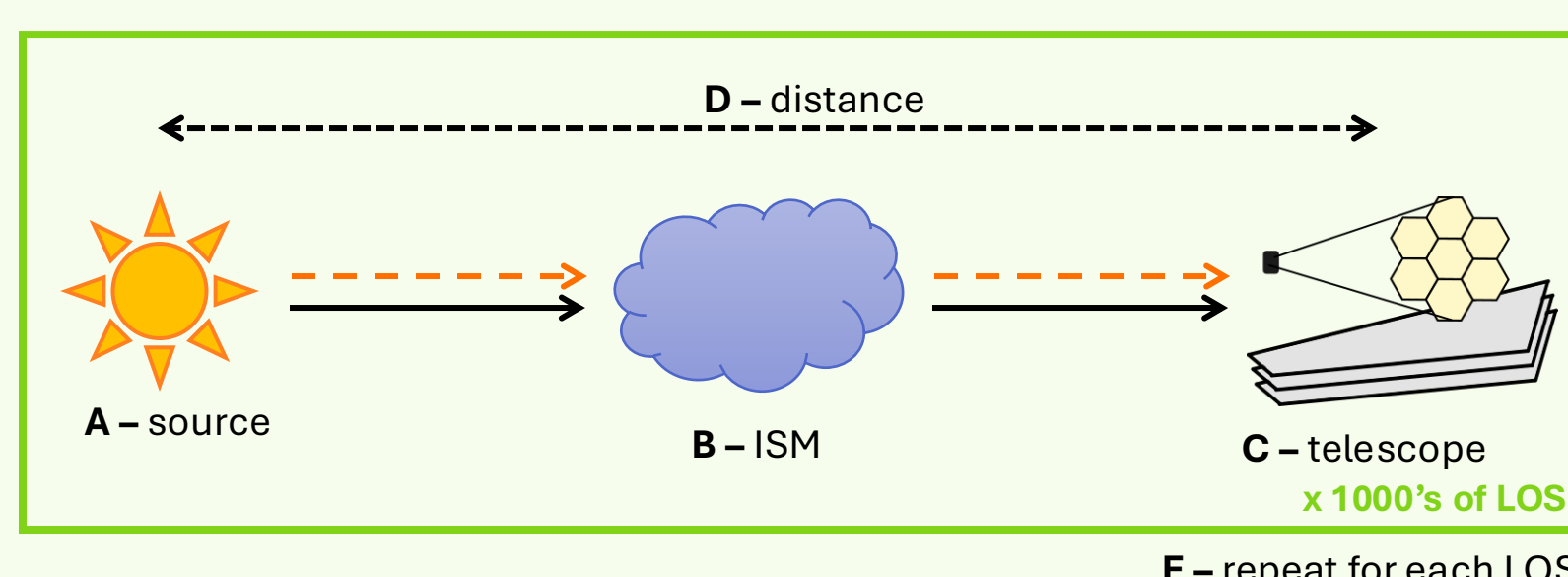
Project aim is to "fit" and "match" these simulations to the observed spectrum of each LOS

MACHINE LEARNING

- Find best fitting continuum given an observation
- Repeat for all LOS (1000's per molecular cloud)

Data intensive problem, exploring different solutions:

- **Nested Sampling:** for many parameters, the prior can be approximated, leading to physics-informed solutions. *Issues:* discrete grid + can be slow and not converge correctly.
- **GPR:** used to interpolate the discrete grid of simulations
- **CNN:** by carefully choosing the convolution layers, I can focus on both small-scale variations (photospheric lines, metallicity, $\log g$ and T_{eff}) as well as large-scale ones (extinction, T_{eff}). *Issues:* may need more training data, few labelled real observations.



CONCLUSIONS

First results show accurate simulations of observations, highlighting the viability of this method to untangle the effects of gas, dust and ice in molecular clouds.

These simulations were done by manual tuning of the parameters, and better match is expected to come from a search/fit algorithm.

FUTURE WORK

- Validate with known stars, using on/off cloud stars and Gaia data
- Analyse and compare different clouds: Cha I, B335, B68, LDN 694-2
- Study how extinction may differ between clouds

